Activity Overview: Research, Build, and Decide

This activity is designed to help you practise **turning data into strategic insight**. You will source a dataset, build a simple interactive dashboard, generate AI-supported insights, and use those insights to inform business decisions. The goal is to understand how AI enhances decision-making, and where human judgment remains essential.

You are encouraged to choose a dataset related to an industry or topic that interests you, such as retail sales, subscription trends, logistics performance, or customer satisfaction.

Tools and Tutorials

You may choose either **Power BI (Free)** or **Tableau Public (Free)** to build your dashboard.

Below are **beginner-friendly tutorials** and setup instructions.

Data Sources (Free)

Choose one dataset from:

- Kaggle: **https://www.kaggle.com/datasets**
- Google Dataset Search: **https://datasetsearch.research.google.com**
- UCI ML Repository: **https://archive.ics.uci.edu/ml/datasets**

Tip: Search terms such as *retail sales*, *e-commerce transactions*, *customer churn*, or *store performance*.

Power BI Free Setup and Tutorial

1. Download Power BI Desktop : **https://powerbi.microsoft.com/desktop**
2. Intro Tutorial (Video):**https://www.youtube.com/watch?v=AGrl-H87pRU**
3. Connecting Your Data to Power BI: File → Get Data → Excel/CSV → Load → Visualizations pane to build visuals

Tableau Public Setup and Tutorial

1. Download Tableau Public: **https://public.tableau.com/app/discover**
2. Intro Tutorial (Video):**https://www.youtube.com/watch?v=TPMlZxRRaBQ**
3. Connect Data → Drag fields into Rows and Columns → Use the Show Me panel to select chart types

Optional: AI Support for Insight Interpretation

You may use ChatGPT or Gemini to generate analytical summaries.

Example prompt:

“Here is a summary of my dataset: [paste description]. Identify trends, performance differences, and potential causes.”

Step 1: Select and Explore Your Dataset

- Select a dataset with at least **500 rows** and multiple variables (e.g., date, region, product category, revenue).
- Open it in Excel or Google Sheets.
- Examine:
    - Which variables drive performance?
    - How data changes over time, region, or category
- Write **3–5 initial observations** about what stands out.

Step 2: Construct Your Dashboard

Using Tableau Public or Power BI:

- Create **at least four** visual elements, such as:
    - Sales or activity trend over time
    - Performance comparison across categories or regions
    - Customer or product segmentation breakdown
    - A summary or KPI panel
- Add **filters or slicers** so the viewer can interact with the dashboard.
- Ensure clarity: visuals should be readable and purposeful.

Step 3: Generate AI-Supported Insights

- Summarise your dashboard in text (2–3 sentences).
- Provide the summary to ChatGPT or Gemini.
- Request diagnostic and forward-looking insight suggestions.

Example prompt:

“Based on my dashboard [paste findings], identify potential causes of performance variation and suggest possible strategic responses.”

Step 4: Data-Driven Decision Simulation

Using the insights you have generated:

- Make **three business recommendations**, such as:
    - Where to invest

    - What to reduce or discontinue
    - Which regions/products to prioritise
    - How staffing or distribution be adjusted
- Each recommendation must reference:
    - A visual pattern in your dashboard
    - An AI-generated insight
    - Your reasoning as the decision-maker

Deliverables

Submit the following:

1. **Dashboard File**
    - Export your final dashboard as a **PDF** showing all key visuals.
2. **Insight Summary (200–300 words)**
    - What the data shows
    - Key patterns and anomalies
    - AI-generated insight highlights
3. **Decision Brief (300–450 words)**
    - Three strategic recommendations
    - Justification using both data and judgment
    - Acknowledgement of uncertainties or limitations

Grading

Marks for each of the criteria will be assigned accordingly:

- 0 - Failed to submit
- 1 - Limited
- 2 - Basic
- 3 - Proficient
- 4 - Advanced
- 5 - Exceptional

The following assessment criteria will be used to grade your submission:

- Effort
- Application
- Examples
- AI for Business insight

Rubric

Criteria	1	2	3	4	5
	Limited	**Basic**	**Proficient**	**Advanced**	**Exceptional**
Effort	Minimal effort apparent, with little attention to detail or engagement with the task.	Shows some effort, but responses may lack polish or thoroughness.	Demonstrates a solid effort, with responses showing attention to detail and engagement with the task.	Displays a strong effort, with responses exhibiting thoroughness, coherence, and depth.	Exemplifies an outstanding effort, with responses displaying exceptional clarity, insight, and completeness.
Application	Fails to apply learned information effectively to the scenario.	Demonstrates basic application of learned information to the scenario, but lacks depth or coherence.	Applies learned information to the scenario with clarity and coherence, demonstrating a solid understanding of its relevance.	Applies learned information skillfully to the scenario, offering insightful analysis and connections.	Applies learned information in a highly sophisticated manner, integrating multiple perspectives and drawing nuanced conclusions.
Examples	Few or no examples provided. Demonstrates limited understanding or application of concepts.	Provides some examples, but they may be simplistic or lack depth. Demonstrates basic understanding of concepts.	Offers several relevant examples demonstrating understanding of concepts. Examples are well-integrated into the response.	Presents numerous detailed examples that vividly illustrate concepts and their applications.	Presents numerous detailed examples that vividly illustrate concepts and their applications.
AI for Business Insight	Shows little understanding of the principles of AI for Business Strategy and fails to connect theories to the scenario.	Displays basic understanding of AI for Business Strategy; makes simple, expected connections to the scenario.	Demonstrates a clear understanding of AI for Business Strategy and makes relevant connections to the scenario, reflecting solid insight.	Exhibits deep understanding of AI for Business Strategy concepts; offers thoughtful, original insights that enhance the scenario analysis.	Demonstrates an understanding of AI for Business while; integrating sophisticated insights that significantly enrich the scenario, reflecting a high level of critical thinking.